Event-Driven Programming

Mr. E. O. Tugbeh
AI Product Engineer, HWP Labs.

About Me

- Bachelor's degree in Mathematics & Computer Science
- Master's degree in Computer Science (Advanced Software Engineering)
- Over 12+ years of experience in **Enterprise Software Development & Training**
- Worked across critical industries such as Banking, Education, Government & Healthcare
- Member of Computer Professionals (Registration Council of Nigeria) [CPN]
- Member of Google Developer Group (GDG) & Student Ambassador 2013/2014
- Microsoft Certified Azure Developer Associate (AZ-204)
- Claude Certified Solutions Architect & Discord Server Community Mentor

Overview

0 Role of JavaScript

1 JavaScript Fundamentals

2 BOM BOM & DOM

3 Events Triggers & Handlers

4 Vibe Coding w/Claude AI

Prerequisites

0

HTML/CSS Fundamentals

1

Charged Laptop

2

Internet Connection

Role of JavaScript

Webpage Interactivity

> JavaScript is an **interpreted**, high-level programming language used to develop interactive webpages.

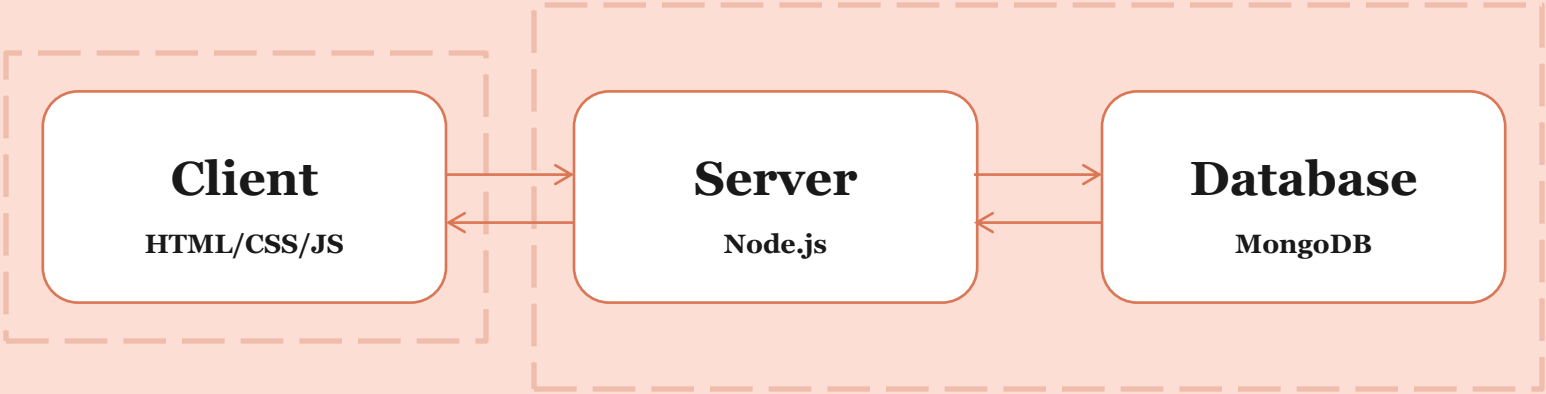
Trivia

- **1995** - JavaScript was created by Brendan Eich (Netscape, *within 10 days*).
- **2009** - Node.js was created by Ryan Dahl as a JS runtime environment to execute JS outside web browsers.
- **2011** - React.js was created by Jordan Walke (Facebook) as an open-source JS library for building single-page applications (SPAs).

Application Areas

- Manipulate HTML & CSS (DOM Elements)
- Handle Webpage Interactions
- Develop APIs/Web Services (Node.js)
- Build Desktop Apps (Electron)
- Build Mobile Apps (React Native)
- Interface with ETH Smart Contracts (Web3.js)

Web App Architecture



Questions

1

JavaScript Fundamentals

Syntax & Semantics

- **Variables & Constants** – Named storage location in memory that can/cannot be changed after declaration (var, const).
- **Data Types** – Defines the type of value to be stored in memory, how much memory it consumes, and what type of operations can be performed on it;
  - Primitives (Number, String, Boolean, null, undefined)
  - Non-Primitives (Array, Object, Date)
- **Operators** – Symbols used to perform computations;
  - Arithmetic & Assignment
  - Comparison, Logical & *Ternary

Variables & Constants

1.1

Example 1

1. // TODO: Compute Student Grade
2. var score = 47;
3. const total = 60;
4. const grade = (score * 100) / total; // 78.33%
5. const passed = score >= (total / 2); // true
6. const remarks = passed ? “Pass” : ”Fail”;
7. console.log(grade, passed, remarks);

Identifiers

1. The name given to a variable is known as an **Identifier**.
2. Identifiers can begin with or contain special characters like $ and _.
3. Numbers cannot be used as the first character of an Identifier.
4. The JavaScript convention for Identifiers is ***camelCase**.
5. Identifiers should be as descriptive as possible. This helps with readability.

Example 2

```
1. // TODO: Compute Person Age
2. var birthYear = 1952; // BAT
3. const thisYear = new Date().getFullYear(); // 2026
4. const age = thisYear - birthYear; // 74
5. const canVote = age >= 18 ? "Sure" : "Nope";
6. console.log(thisYear, age, canVote);
```

Comments

- Comments are primarily used to **document** code snippets.
- Comments can also be used to ignore code snippets during **debugging**.
- Inline comments begin with // **...**
- Multi-line comments are wrapped within /* **...** */

Data Types

1.2

Array (*List)

- An Array is an ordered *list* of values known as **elements**.
- Arrays can store elements of different data types (numbers, strings, booleans, etc).
- Array elements are accessed based on their **index**.
- The first element in an Array is located at **index[0]**.
- The last element in an Array is located at **index[n−1]**, where *n* is the Array size.
- The size of an Array in JavaScript changes as elements are added or removed.
- A nested Array is called a **Matrix**, i.e. an Array of Arrays.

Example 1

```
1.  // TODO: Process Jamb Scores
2.  const jambScores = [56, 42, 55, 60]; // ENG, MTH, PHY, ECO
3.  var size = jambScores.length; // 4
4.  var english = jambScores[0]; // 56
5.  var economics = jambScores[size - 1]; // 60
6.  var scored_56 = jambScores.includes(56); // false
7.  var scored_76 = jambScores.includes(76); // true
8.  var removedLastValue = jambScores.pop(); // [56, 42, 55], 60
9.  var newArraySize = jambScores.push(55); // [56, 42, 55, 55], 4
10. var removedFirstValue = jambScores.shift(); // [42, 55, 55], 56
11. var newArraySize = jambScores.unshift(76); // [76, 42, 55, 55], 4
12. var sortedScoresAsc = jambScores.toSorted((a, b) => a - b); // [42, 55, 55, 76];
13. var sortedScoresDesc = jambScores.toSorted((a, b) => b - a); // [76, 55, 55, 42];
14. var totalScore = jambScores.reduce((total, value) => total + value, 0); // 228
15. var minScore = Math.min(...jambScores); // 42
16. var maxScore = Math.max(...jambScores); // 76
17. var verifyDatatype = Array.isArray(jambScores); // true
18. console.log(jambScores, totalScore, sortedScoresAsc, sortedScoresDesc);
```

Object (*HashMap)

- An Object is a record of **key:value** pairs called **properties**.
- Objects can store properties of different data types (numbers, strings, booleans, etc).
- Object properties are accessed based on their **key**, either using the "dot-notation" **record.key** (preferred) or the "index-notation" **record[key]** (dynamic).
- **Object.keys(record)** returns Object keys as Array, **Object.values(record)** also returns Object values as Array & **Object.entries(record)** returns Object properties as *Matrix*.
- **JSON.stringify(record)** returns Object properties as a *serialized* string.
- A nested Object is called a **Tree**, i.e. an Object of Objects.
- An "Array of Objects" is called a **Collection**.

Example 2

```
1.  // TODO: Process Jamb Scores
2.  const jambScores = { ENG: 56, MAT: 42, PHY: 55, ECO: 60 };
3.  var size = Object.keys(jambScores).length; // 4
4.  var english = jambScores["ENG"]; // 56 (index-notation)
5.  var economics = jambScores.ECO; // 60 (dot-notation preferred)

6.  jambScores.PHY = 71 // {ENG: 56, MAT: 42, PHY: 71, ECO: 60}
7.  jambScores.CHE = 35 // {ENG: 56, MAT: 42, PHY: 71, ECO: 60, CHE: 35}
8.  delete jambScores.CHE; // {ENG: 56, MAT: 42, PHY: 71, ECO: 60}
9.  var wroteChemistry = Object.keys(jambScores).includes("CHE"); // god forbid!
10. var scored_56 = Object.values(jambScores).includes(56); // true
11. var totalScore = Object.values(jambScores).reduce((total, value) => total + value, 0); // 229
12. var minScore = Math.min(...Object.values(jambScores)); // 42
13. var maxScore = Math.max(...Object.values(jambScores)); // 71
14. var verifyDatatype = typeof jambScores === 'object'; // true
15. console.log(jambScores, totalScore, wroteChemistry, verifyDatatype);
```

Date

- Date is an in-built JavaScript **module** for processing date and time.
- A Date instance can be created with **new Date()**, which returns the current browser/server timestamp. And also with **new Date("YYYY-MM-DD")** OR **new Date("YYYY-MM-DD HH:MM:SS")**
- **new Date().toString()** returns **"Fri Jul 03 2026 04:00:12 GMT+0100 (West Africa Time)"**.
- **new Date().toISOString()** returns **"1970-01-01T00:00:00.000Z"** - The standard format for storing timestamps.
- **Date.now()** returns the total number of milliseconds since **Unix Epoch Time** (the beginning of time computations).
- A nested Date is called a **Calendar**, i.e. a Date of Dates? Na lie o! ☺

Example 3

```
1.  // TODO: Format datetime as Sun, 8 Jul 2026 | 10:00 AM
2.  const dateObj = new Date("2020-10-20 15:00:00"); // #EndSARS Protest
3.  const dayOfWeekIndex = dateObj.getDay(); // 0-6
4.  const dayOfMonth = dateObj.getDate(); // 1-31
5.  const monthIndex = dateObj.getMonth(); // 0-11
6.  const year = dateObj.getFullYear(); // 2026
7.  const hourIndex = dateObj.getHours(); // 0-23
8.  const minuteIndex = dateObj.getMinutes(); // 0-59
9.
    const days = ["Sun","Mon","Tue","Wed","Thu","Fri","Sat"];
10. const months = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"];
11. const meridiem = hourIndex < 12 ? "AM" : "PM";
12. const hours_f = hourIndex % 12 || 12; // 1-12
13. const minutes_f = minuteIndex.toString().padStart(2, "0"); // 00-59
14. const date_f = `${days[dayOfWeekIndex]}, ${dayOfMonth} ${months[monthIndex]} ${year}`;
15. const time_f = `${hours_f}:${minutes_f} ${meridiem}`;
16. const output = `${date_f} | ${time_f}`;
17. console.log(dateObj.toString(), output);
```

1.3

Operators

Arithmetic

Oper	Name / Link	Example
+	Addition	x = y + 2
-	Subtraction	x = y - 2
*	Multiplication	x=y*2
/	Division	x = y / 2
**	Exponentiation	x = y ** 2
%	Remainder	x = y % 2
++	Increment	x = y++
--	Decrement	x = y--

Assignment

Oper	Name / Link	Example	Same As
=	Simple	x = y	x = y
+=	Add	x += y	x = x + y
-=	Subtract	x -= y	x = x - y
*=	Multiply	x *= y	x = x * y
/=	Divide	x /= y	x = x / y
%=	Remainder	x %= y	x = x % y

Comparison

Oper	Name / Link	Comparing	Returns
==	Equal to	x == 8	false
==	Equal to	x == 5	true
===	Strict Equal	x === "5"	false
===	Strict Equal	x === 5	true
!=	Not Equal	x != 8	true
!==	Strict not Equal	x !== "5"	true
!==	Strict not Equal	x !== 5	false
>	Greater than	x > 8	false
<	Less than	x < 8	true
>=	Greater or Equal	x >= 8	false
<=	Less or Equal	x <= 8	*true*

Logical

Oper	Name / Link	Example
&&	AND	(x < 10 && y > 1)
||	OR	(x === 5 || y === 5)
!	NOT	!(x === y)
??	Nullish Coalescing	x ?? y

Questions

2

BOM BOM & DOM

Browser/Document Object Model

3

Events Triggers & Handlers

4

Vibe Coding w/Claude AI

>_

[attach-screenshot] generate static html5 semantic tags and css3 in single file

Exercise 1

- **Northwind AI** is an AI Automation and Data Analytics service, provided by HWP Labs.
- **Northwind AI Podcast** is a platform that promotes tech startups, and application areas of AI within and outside the tech industry.
- Northwind AI Podcast holds LIVE on X Spaces, on Fridays & Sundays at 8pm (WAT).
- **Tech Stack:** TypeScript (Next.js), Python (FastAPI), PostgreSQL (Supabase) & Vercel
- **TODO**: "Vibe code" Northwind AI Podcast homepage using Claude AI, and implement *load*, *input* & *click* events with JavaScript.

Live Demo

https://github.com/2gbeh/src/tree/main/event-driven-programming/northwind-ai

Exercise 2

- **OPay** is a FinTech **super-app** founded in 2018, and acquired by Opera Software.
- OPay operates across Nigeria, Egypt, Pakistan and other markets offering Mobile Money (payments, transfers, loans, utility bills), ATM Cards, POS Terminals, Ride-hailing, and Food Delivery services.
- OPay reported serving 45m+ Users, and 1m+ Merchants & Businesses in Nigeria.
- **Tech Stack:** React Native, Kotlin, Swift, Node.js, PostgreSQL, Redis, Kafka & AWS
- **TODO**: "Vibe code" OPay App transfer screen using Claude AI, and implement *input* & *click* events with JavaScript.

Live Demo

https://github.com/2gbeh/src/tree/main/event-driven-programming/opay

Exercise 3

- **ExpoCBT** is an open-source CBT **WebApp** developed by E.O. Tugbeh & H.M. Oshomoh.
- ExpoCBT will be **"built in public"** on **TikTok LIVE**, alongside students of the Dept. of Software Engineering, Obafemi Awolowo University (OAU).
- Active participants will be added on GitHub as Contributors and assigned Issues.
- **Tech Stack:** TypeScript (Next.js), PostgreSQL (Supabase) & Vercel
- **TODO**: "Vibe code" ExpoCBT using Claude AI, implement *load*, *input* & *click* events with JavaScript, and deploy on a Live production env using Vercel.

Live Demo

https://github.com/hwp-labs/expo-cbt

NOTE: Create a GitHub account and **[Star]** the repository to be added as a Contributor!

Questions

Resources

1. Uvietobore, T. (2026). [WATCH] SEN106/SEN216 - Introduction to Web Technologies: https://youtube.com/watch?v=cGoyAmGwDog,
2. Tugbeh, E.O. (2026). [PDF] Event-Driven Programming: https://github.com/2gbeh/src/blob/main/event-driven-programming/slide.pdf
3. Tugbeh, E.O. (2026). [ZIP] Source Code Repository: https://github.com/2gbeh/src/blob/main/event-driven-programming/lab.zip
4. Tugbeh, E.O. (2026). Code Snippets Markdown File: https://github.com/2gbeh/src/blob/main/event-driven-programming
5. OneCompiler JavaScript Online Interpreter - https://onecompiler.com/javascript
6. Tugbeh, E.O., Oshomoh, H.M., & Aiyeduyoni S. (2026). Northwind AI Podcast E19 - Statement of Work: Government Subcontractors: https://x.com/i/spaces/1wGWjjElmNNKQ
7. Join Northwind AI Community: https://chat.whatsapp.com/HajjJPJ2HkEorTTITv2FGm
8. Support Northwind AI: https://support.northwindai.org
9. W3Schools – JavaScript Tutorial: https://w3schools.com/js/default.asp
10. W3Schools – Introduction to Programming: https://w3schools.com/programming/index.php

Let's Connect

Email	etugbeh@outlook.com
Telephone	(+234) 81 6996 0927
LinkedIn	/in/2gbeh
GitHub	/2gbeh
Twitter/X	/@2gbeh
TikTok	/@2gbeh
Website	https://northwindai.org
Podcast	https://podcast.northwindai.org
Community	https://chat.whatsapp.com/HajjJPJ2HkEorTTITv2FGm

Thank You!